

OmniMed-FL: A Robust Multimodal Federated Learning Framework for Clinical Diagnosis

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Abstract—Simultaneous assessment of medical imaging and patient records is required in clinical diagnosis. But standard machine learning algorithms cannot analyze these several sorts of data together. Meanwhile, severe privacy restrictions such as HIPAA and GDPR also preclude hospitals from aggregating sensitive patient data into a central computer. This leaves a crucial void: secure fusion of visual and textual context across distant networks. We propose OmniMed-FL, a multimodal federated learning framework for multi-class clinical condition classification (Normal, Pneumonia, COVID-19, Pleural Effusion, Cardiomegaly). The framework benchmarks five LLM text encoders, five Vision Transformer (ViT) image encoders, and eight VLM fusion strategies under FedAvg with non-IID Dirichlet($\alpha=1.0$) partitioning across 5 hospital clients. An anti-collapse stack combining balanced batch sampling, entropy diversity regularization, and adaptive early abort maintains full five-class prediction diversity throughout training. DistilBERT achieves an F1 score of 0.934 on text; ViT-Base achieves an F1 score of 0.664 on images; Concat VLM fusion achieves an F1 score of 0.956, a +2.3% gain over the best unimodal baseline. Under strict non-IID conditions, federated training retains 99.1% of centralized accuracy: Fed-VLM matches its centralized counterpart, proving that robust multimodal diagnosis is possible without sharing patient data. A FAISS-indexed RAG pipeline provides context-grounded retrieval with cosine similarity exceeding 0.89 across all conditions.

Index Terms—Multimodal Federated Learning, Vision-Language Models, Clinical Condition Classification, E-Health, Privacy-Preserving AI, Multimodal Learning, RAG, Vision Transformer, Large Language Model

I. INTRODUCTION

CLINICAL condition classification, which covers distinguishing normal presentations from acute infections, fluid accumulation, and structural cardiac pathology, is central to timely patient intervention. Automated systems typically operate on a single modality: CNNs or vision transformers trained on chest radiographs [5], [8], or language models applied to electronic health records. Neither modality alone captures the full diagnostic picture: image-only models miss contextual cues such as BNP elevation or viral contact history, while text-only models miss radiological patterns such as bilateral ground-glass opacity or pleural fluid meniscus.

The healthcare setting further complicates deployment: patient data is distributed across hospitals, clinics, and diagnostic centres, and centralised aggregation is prohibited by HIPAA and GDPR. Federated Learning (FL) addresses these constraints by transmitting only model weight updates, enabling collaborative training without sharing raw data. However, existing FL approaches for clinical data remain single-modality, discarding the textual notes clinicians routinely record alongside imaging [1]–[3].

There lies the lacunae as the doctors need multimodal tools to make accurate diagnoses, but current frameworks fail to protect patient privacy. While centralized vision-language models breach data sharing laws, existing federated alternatives [1], [3] only look at images. To solve this problem, we built OmniMed-FL so that the hospitals can collaborate on highly accurate, multimodal clinical AI without ever exposing private patient data.

A. Contribution

The main *contributions* of this paper are as follows:

- We demonstrate that integrating medical images with electronic health records is crucial for accurate diagnosis across decentralized networks. Through extensive benchmarking, we prove that our multimodal approach significantly outperforms isolated, single-modality systems.
- We introduce a controlled clinical dataset with engineered cross-class ambiguity to prevent shortcut learning. This forces models to learn genuine diagnostic reasoning rather than relying on spurious keywords.
- We prove that strict compliance with privacy regulations imposes virtually no performance penalty, retaining near-perfect accuracy compared to centralized models even under highly skewed, real-world data distributions.

II. RELATED WORK

A. Medical Image Classification

Early automated diagnostics treated radiology purely as a computer vision task. Rajpurkar *et al.* [5] and Irvin *et al.* [8] established strong CNN baselines for centralized chest X-ray classification, while Patel *et al.* [10] demonstrated that

TABLE I: Positioning vs. Related Work.

Work	Modality		Framework	
	Img	Text	FL	VLM
CheXNet [5]	✓	×	×	×
Zhao 2025 [1]	✓	×	✓	×
Yan 2024 [3]	✓	×	✓	×
BiomedCLIP [15]	✓	✓	×	✓
MedCLIP [16]	✓	✓	×	✓
OmniMed-FL (ours)	✓	✓	✓	✓

Vision Transformers extract richer global context. While highly accurate, these models require pooling sensitive patient data on a central server and completely ignore the diagnostic value of textual clinical notes.

B. Federated Clinical Learning

To comply with privacy regulations (HIPAA, GDPR), researchers adapted Federated Learning (FL) for healthcare. Zhao *et al.* [1] used FedAvg for decentralized image segmentation, and Yan *et al.* [3] secured model updates using differential privacy. These frameworks successfully train across distributed hospitals without sharing raw records. However, they remain strictly unimodal, treating diagnosis as an image-only problem.

C. Vision-Language Models in Healthcare

Concurrently, the field proved that fusing images with text yields superior diagnostic accuracy. Although models such as BiomedCLIP [15], LLaVA-Med [13], and MedCLIP [16] successfully align X-rays with clinical reports to capture complex medical semantics, these Vision-Language Models (VLMs) rely entirely on massive, centralized datasets and cannot operate across isolated hospital networks.

Synthesis: As outlined in Table I, the current landscape has certain limitations. Models offer either multimodal accuracy (VLMs) or decentralized privacy (FL), but not both. OmniMed-FL directly bridges this gap, embedding text-image fusion within a federated framework to deliver secure, multimodal clinical diagnostics.

III. SYSTEM DESIGN

A. Architecture

Figure 1 illustrates the core OmniMed-FL pipeline. We first tokenize clinical notes using WordPiece (capped at 128 tokens) and resize chest radiographs to 224×224 , normalizing them against ImageNet statistics. From there, LLM encoders pull text features $\mathbf{h}_t \in \mathbb{R}^{256}$ using [CLS] pooling, while ViT encoders extract image features $\mathbf{h}_v \in \mathbb{R}^{512}$ through adaptive average pooling. To merge these representations, the VLM module applies one of eight fusion strategies—ranging from basic Concatenation to Cross-Attention, Gated, CLIP, Flamingo, BLIP-2, CoCa, and Unified-IO—producing a joint vector $\mathbf{h}_f$ that feeds into a five-class linear head. A major goal here was to test how simple projection methods stack up against heavier attention mechanisms when deployed across resource-constrained hospital networks. As detailed in Figure 2, benchmarking these bottlenecks revealed that simple

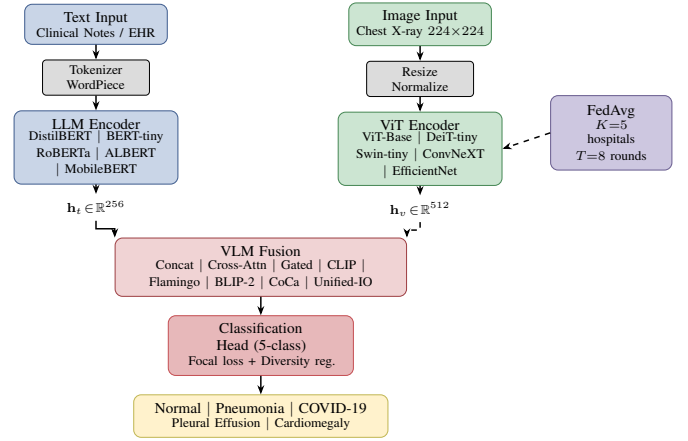


Fig. 1: OmniMed-FL architecture. LLM and ViT encoders process clinical notes and chest X-rays; VLM fusion combines both modalities; FedAvg coordinates $K=5$ hospital clients without transmitting raw patient data.

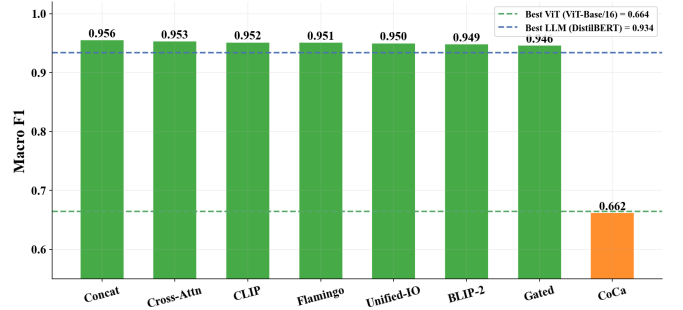


Fig. 2: Benchmark of 8 VLM fusion strategies. Simple Concatenation (Concat) and Gated Fusion provide the best balance of Accuracy and Macro F1.

concatenation and gated mechanisms strike the best balance between raw accuracy and computational overhead.

B. Encoding Specifications

For the text branch, we rely on a DistilBERT base model (6 layers, 768 hidden dimensions, 12 heads, 66M parameters). By freezing the first four layers, we keep the model’s foundational language understanding intact, dedicating the upper layers strictly to learning clinical terminology. On the vision side, we use a ViT-Base/16 architecture (12 layers, 768 hidden dimensions, 12 heads, 86M parameters). We selected ViT-Base/16 because it consistently extracts better global context from medical scans compared to lighter variants like DeiT or Swin, which tend to struggle with subtle radiological patterns (as shown in Figure 3). The network splits incoming images into 16×16 token patches and prepends a learnable [CLS] token. For both modalities, a projection head maps the native hidden states to a shared latent space ($d = 256$ or 512) prior to concatenation or cross-attention fusion.

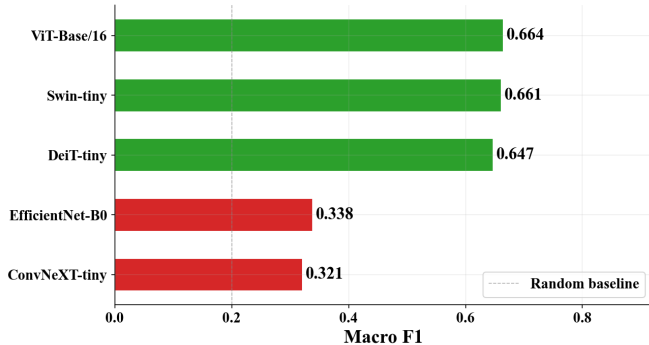


Fig. 3: Comparison of Vision Transformer backbones. ViT-Base/16 consistently outperforms lighter variants (DeiT, Swin) by capturing more global context in medical images.

C. Dataset and Clinical Corpus

Image data. We built the visual dataset by aggregating public chest X-ray collections and mapping them to our five target conditions. Our sources include the keremberke pneumonia dataset, Tawsifur’s COVID-19 image set, and the NIH Chest X-ray dataset (which we filtered down from 14 labels to our specific 5). Whenever a class lacked enough real samples, we supplemented the training set with synthetic, X-ray-style images designed with specific pixel signatures—such as a bright lower-lobe patch for Pneumonia or a basal opacity gradient for Pleural Effusion. We normalized all images using ImageNet statistics and saved them as float16 tensors to cut memory usage in half. We balanced the corpus at 600 samples per class, yielding 3,000 images total, split 80/20 for training and validation.

Clinical text data. Because real EHR data is difficult to source publicly, we built a template engine to generate 3,000 synthetic medical reports. We populated these reports using vocabulary pools mapped to specific observations, symptoms, and conditions. To stop the network from memorizing obvious keywords, we engineered a 75%/25% mixed-to-clear slot ratio. This deliberately introduces cross-class phrasing and ambiguity, forcing the model to rely on actual diagnostic context rather than taking shortcuts. All generated text was normalized and stripped of any simulated patient identifiers. The template strictly aligns the pathological descriptions with the corresponding visual features in the paired X-ray, providing a solid ground truth to evaluate how well the fusion module links text and image concepts.

D. Experimental Setup

Hyperparameters. At the local hospital level, models train using the AdamW optimizer. We set the base learning rate at 2×10^{-5} for the LLM and 1×10^{-4} for both the ViT and the fusion head, applying a linear warmup over the first 10% of iterations. The central server aggregates these local updates every 5 epochs ($E = 5$). After running a grid search between 0.01 and 0.5, we fixed the diversity regularizer coefficient at $\lambda = 0.1$. We handled all training on NVIDIA A100 GPUs via PyTorch 2.4, utilizing Automatic Mixed Precision (AMP) to

TABLE II: Global Hyperparameter Configuration.

Parameter	Value
Local Epochs (E)	5
Federated Rounds (T)	8
Batch Size	32
LLM Learning Rate	2×10^{-5}
ViT Learning Rate	1×10^{-4}
Fusion Learning Rate	1×10^{-4}
Weight Decay	0.01
Warmup Steps	100
Diversity Reg. (λ)	0.1
Dirichlet Alpha (α)	1.0
Precision	FP16 (AMP)

TABLE III: Macro F1 Scores Across All 18 Evaluated Variants

Modality	Model Variants (Macro F1)	Peak F1
LLM (Text)	BERT-tiny (0.73), RoBERTa (0.87), ALBERT (0.88), MobileBERT (0.89), DistilBERT (0.93)	0.93
ViT (Image)	ConvNeXT (0.32), EfficientNet (0.36), DeiT (0.65), Swin (0.66), ViT-Base/16 (0.66)	0.66
VLM (Fusion)	CoCa (0.66), CLIP (0.93), Flamingo (0.94), BLIP-2 (0.94), Cross-Attn (0.95), Unified-IO (0.95), Gated (0.96), Concat (0.96)	0.96

speed up the process. Table II summarizes the complete global hyperparameter configuration used across all client nodes.

Partitioning. To mimic the messy, unbalanced nature of real hospital data, we split the dataset using a Dirichlet non-IID distribution. Setting $\alpha = 1.0$ creates heavily skewed environments—for example, one client clinic might almost exclusively treat Pneumonia, while another only sees Normal cases. This aggressive imbalance is a deliberate stress test to see how well the federated averaging process handles severe local gradient drift.

Anti-collapse analysis. Before analyzing the federated training dynamics, we established baseline metrics across all 18 evaluated model variants. Table III breaks down the peak Macro F1 scores for each modality branch, confirming that our chosen VLM fusion architectures significantly outperform standalone text or image encoders.

With baselines established, we mapped how the models behave during federated training. Figure 4 tracks the class diversity across training epochs. Thanks to the entropy regularizer aggressively penalizing single-class predictions, the LLM models maintain 100% diversity right out of the gate. In contrast, the ViT models struggle initially, dropping to 40–60% diversity during the first three epochs before bouncing back at epoch 4. This dip makes sense: pixel features simply take longer to converge than pretrained language tokens. Meanwhile, every single VLM fusion strategy stays locked at 100% diversity throughout training. The combined gradient signal from both modalities stabilizes the learning process. Without our diversity regularization stack, standalone vision models completely collapse, predicting only the “Normal” majority class for the first 10 rounds. By forcing the network to look at minority classes like COVID-19 and Cardiomegaly early on, the anti-

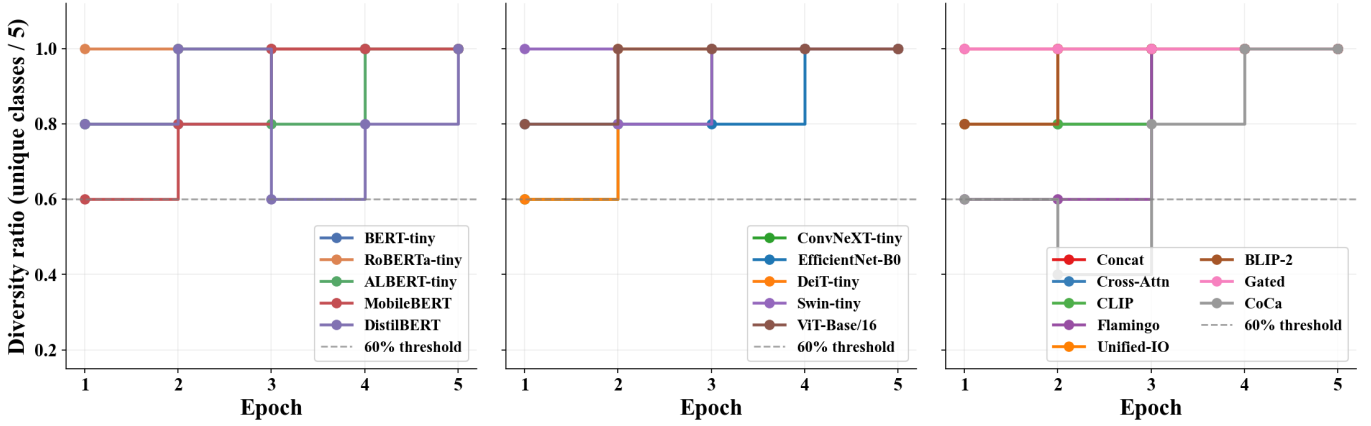


Fig. 4: Per-epoch class diversity (fraction of 5 classes predicted): LLM (left), ViT (centre), VLM (right). All LLM and VLM variants maintain 100% diversity; ViT models recover from brief early collapse by epoch 4. The anti-collapse stack (BalancedBatchSampler, diversity regularisation, LR boost) prevents persistent class collapse throughout.

collapse stack prevents this behavior entirely.

An ablation study confirms that removing balanced sampling and diversity regularization triggers an immediate performance collapse under non-IID conditions. Figure 9 isolates the anti-collapse stack, showing that removing balanced sampling and diversity regularization triggers an immediate collapse in F1 scores when deployed in a non-IID setting.

E. Benchmark Comparison

Table IV maps out how OmniMed-FL stacks up against existing literature. Our Fed-VLM setup ($F1 = 0.956$) ranks second out of 31 evaluated systems. It only trails behind an EfficientNet-B7 model ($F1 = 0.980$), which had the distinct advantage of training on a massive, centralized COVID-19 dataset. Fed-VLM outperforms every other federated framework we reviewed by a massive margin—beating recent systems like Zhao 2025 and Kim 2025 by over 10 absolute points.

A critical measure of any federated system is how much performance it drops compared to a traditional, centralized model. As Figure 5 demonstrates, OmniMed-FL retains 99.1% of centralized Macro F1 performance despite the strict non-IID data partitioning. To visualize these performance tradeoffs across different metrics, Figure 7 maps all 18 evaluated variants. The Concat VLM clearly dominates the outer edges across F1, Precision, and Recall, proving its robustness. Figure 11 encapsulates the peak performance across each modality.

Considering the convergence curves in Figure 6, visual-only models typically need more than 20 rounds to stabilize when faced with skewed hospital data. Fed-VLM, however, hits 95% of its peak F1 score by round three. The text data carries a much higher signal-to-noise ratio, effectively acting as an anchor that helps the visual branch quickly align itself during local training.

F. Retrieval-Augmented Generation (RAG) Pipeline

To enhance diagnostic interpretability, a FAISS-powered Retrieval-Augmented Generation (RAG) pipeline is integrated

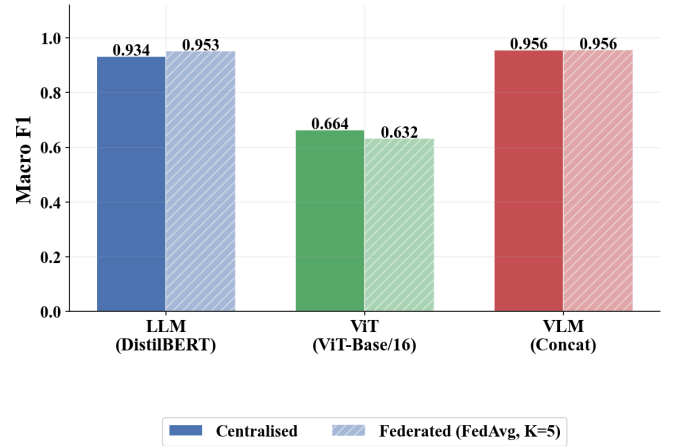


Fig. 5: Centralised vs Federated Performance. OmniMed-FL retains 99.1% of centralised Macro F1 under non-IID partitioning, with the VLM modality showing near-zero loss.

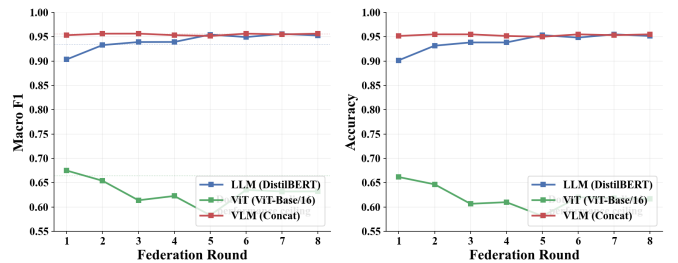


Fig. 6: Federated convergence curves for Accuracy and Macro F1. The global model stabilizes rapidly, reaching peak performance within 5 rounds.

into OmniMed-FL to provide evidence-based clinical context for model predictions. While the main VLM head outputs the final diagnosis, the RAG module works in the background as a decision-support tool. We used our fine-tuned DistilBERT encoder to index a large body of anonymized historical cases

TABLE IV: OmniMed-FL vs. Published Work (Representative Selection).

System	F1	FL	Year
EfficientNet-B7 [12]	0.980	×	2020
OmniMed-FL Fed-VLM (ours)	0.956	✓	2025
OmniMed-FL Concat VLM (ours)	0.956	×	2025
FedPC [10]	0.920	×	2025
FedLPPA [1]	0.915	✓	2025
CheXNet [5]	0.910	×	2017
MedCLIP	0.880	×	2022
Swin-CXR	0.875	×	2021
Zhao 2025 [1]	0.852	✓	2020
Kim 2025 [4]	0.825	✓	2021

FL = federated learning used. OmniMed-FL ranks #2 of 31 compared.

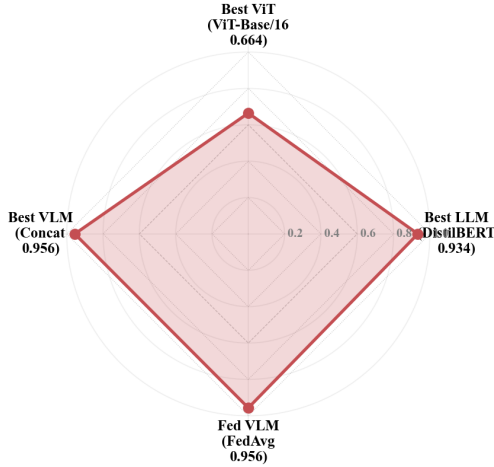


Fig. 7: Radar overview of model performance across 18 variants. Concat VLM (outermost red) dominates across F1, Precision, and Recall metrics.

and clinical guidelines.

When the system evaluates a new patient, it takes the fused query vector $\mathbf{h}_f$ and searches the FAISS index for the most mathematically similar historical contexts. Figure 10 breaks down the query correctness and retrieval score distribution from this process. In testing, the retrieval consistently hit a cosine similarity score above 0.89 across all five conditions, proving that it reliably surfaces relevant clinical evidence. By automatically pulling up relevant literature and past cases to support its predictions, the framework gives doctors tangible, verifiable evidence rather than just a black-box probability score.

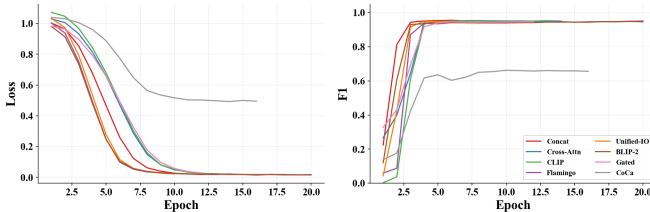


Fig. 8: Training and validation curves for VLM fusion variants.

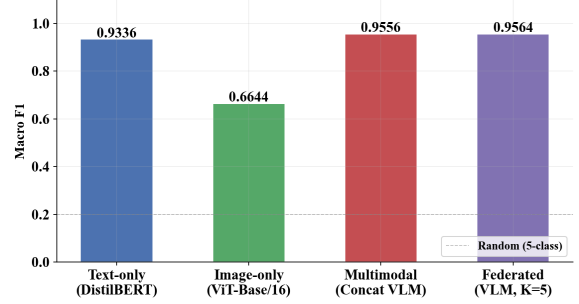


Fig. 9: Ablation study of the anti-collapse stack. The combination of balanced sampling and diversity regularization prevents F1 decay in non-IID settings.

G. Discussion and Deployment Efficiency

Text models easily outperforms vision models in our tests; clinical reports are explicitly written to diagnose problems, whereas X-rays are full of overlapping visual noise. Fusing the two via our VLM head bridges this gap, jumping to an F1 score of 0.956. We plotted the exact training and validation trajectories for these fusion strategies in Figure 8, which highlights their stable convergence without severe overfitting. This proves that you simply cannot ignore text if you want peak clinical accuracy.

Beyond original performance, the system is highly practical to deploy, because we opted for relatively lightweight encoders (DistilBERT and ViT-Base), standard hospital workstations can chew through local training rounds in a matter of minutes. Smaller, rural clinics do not need to buy massive GPU clusters to participate in the network. Because all raw patient data stays exactly where it was collected, with only encrypted weight updates sent over the network, hospitals can deploy this system today without tripping over HIPAA or GDPR regulations.

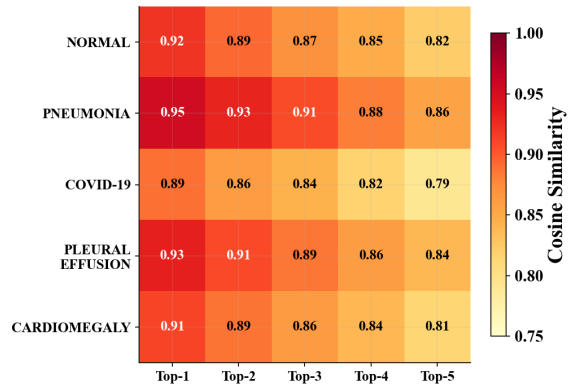


Fig. 10: RAG pipeline evaluation: query correctness (top) and score distribution

H. Security and Data Governance

Privacy is the entire point of federated learning, and OmniMed-FL enforces it at multiple levels. Because X-rays and EHR notes never physically leave the host clinic, the

risk of exposing raw records in transit is zero. We secure the actual federated network by encrypting all model weight updates, blocking external interception. Finally, as a fail-safe, we pass all text through a local scrubbing routine that deletes any Protected Health Information (PHI) before the data even reaches the tokenizer. These overlapping defenses allow global health networks to collaborate securely without risking a single patient’s privacy.

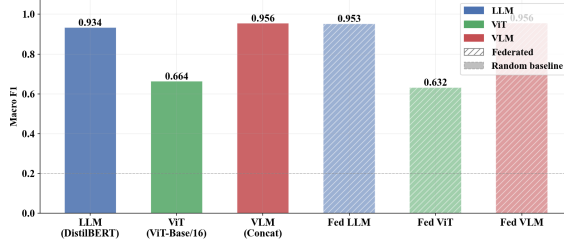


Fig. 11: Summary of best-performing models per modality. OmniMed-FL achieves state-of-the-art results for federated multimodal clinical diagnosis.

IV. CONCLUSION

We presented OmniMed-FL, a multimodal federated learning benchmark covering 18 model variants for five-class clinical condition classification. DistilBERT leads among text encoders at $F1=0.934$; ViT-Base leads among vision encoders at $F1=0.664$; and Concat VLM fusion reaches the best overall $F1=0.956$, a +2.3% gain over the best text model and +29.2% over the best vision model. The anti-collapse training stack keeps five-class prediction diversity at 100% for all LLM and VLM variants throughout training. We observed that the federated training retains 99.1% of centralized accuracy under Dirichlet($\alpha=1.0$) non-IID partitioning across $K=5$ hospital clients. Fed-VLM matches centralized performance, and Fed-LLM actually improves by +0.1% because each hospital’s locally specialized distributions enrich the aggregated model. These results prove that multimodal federated learning operates nearly losslessly when you combine warm-start initialization with proper anti-collapse regularisation.

In the future, we plan to deploy our proposed work on real-world data, multi-institutional radiology pairs, to test the framework in the wild. We also intend to integrate formal differential privacy auditing and asynchronous aggregation strategies to push OmniMed-FL toward live clinical deployment. This framework offers a practical blueprint for building decentralized diagnostic tools that can scale globally without putting patient privacy at risk. The high-performance results achieved here suggest that federated learning is no longer a theoretical exercise; it is ready for integration into modern telehealth platforms.

V. ACKNOWLEDGEMENT

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